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EXPERIENCE REPORT

Learning Health Systems

Developing a data visualization tool for adults with disorders of consciousness: Qualitative analysis of user perspectives

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Abstract

Introduction: We report on the process of using a learning health systems (LHS) approach to design a data visualization dashboard to monitor the rehabilitation progress of patients with disorders of consciousness (DoC) after severe brain injury.

Methods: Our team conducted a qualitative study using informational interviews with key informants to understand informational needs and priorities for the dashboard from the perspectives of rehabilitation therapists, family members of patients with DoC, and third-party payors. We used a thematic survey approach to organize the findings with the following categories: (a) how the dashboard will be used; (b) content to be displayed; (c) organization and design of content; and (d) technical requirements. We used an iterative process to develop the dashboard, with multiple opportunities for stakeholder feedback.

Results: Seven people participated in informational interviews ($n = 2$ rehabilitation therapists; $n = 2$ family members; $n = 3$ third-party payor representatives). The primary intended use of the dashboard is communication and facilitation of shared understanding across clinical teams, rehabilitation teams, and patients' families, and between payors and facilities. Desired content includes core metrics applied by the DoC program for diagnosis and monitoring. There is a high priority for making the display easily understandable and interpretable. Technical requirements include the ability to pull data for display from existing items in the electronic health record to minimize additional burden on therapists. User feedback on the design resulted in a streamlined main screen, with additional detail accessible by clicking into each assessment.

Conclusions: In the unique case of patients with DoC, who cannot speak for themselves, effective communication among rehabilitation clinicians, family members or care partners, and third-party payors is highly important for optimal care. The key benefit of using an LHS approach is positioning the team to proactively design the dashboard to account for the needs and preferences of different end users.

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KEYWORDS
consciousness disorders, data visualization, learning health system, rehabilitation, user-centered design

1 | INTRODUCTION

Data visualization is the representation of information in a chart, diagram, picture, or other pictorial format.[1,2] In healthcare, data visualization may utilize 'dashboards', at-a-glance summaries that convey a variety of patient details in a single location. Dashboard designs often incorporate tools like line graphs and histograms to show changes over time, and color coding to indicate target values. In many cases, dashboards are integrated with electronic health record (EHR) systems to display discrete data elements that are collected as part of routine care. Dashboards can facilitate and track patient progress over time, reduce chart review time for healthcare providers, and facilitate communication between patients and healthcare providers.[3] They can also prompt desired behaviors by drawing attention to measures outside of target ranges. For example, a line graph showing an increase in systolic blood pressure over time may prompt a discussion between patient and provider about medications and lifestyle changes to reduce hypertension.[4]

Healthcare organizations that aspire to become learning health systems (LHSs) aim to systematically use practice-generated data to develop evidence about usual care that can then inform practice improvements.[5] Friedman's model of the continuous learning cycle in an LHS focuses on capturing data about current practice (performance to data), using the data to identify strategies to improve processes or outcomes (data to knowledge), and applying this knowledge, as well as other evidence, to intervene (knowledge to performance).[5] Inclusion of patients and other stakeholders in this iterative improvement process is a cornerstone of the LHS model; thus, dashboard development in an LHS often includes patients and other intended users such as providers and specialists or beneficiaries in the process.[6] The literature provides examples of various levels of patient and frontline provider engagement in data visualization projects ranging from soliciting feedback on a design created by an informatics team to proactive input on decisions about visual characteristics, organization, and color schemes.[7–11] The LHS approach contrasts with more traditional "top down" approaches in which tools such as dashboards are designed by technical experts to meet the needs of leaders or administrators, and presented to end users and beneficiaries for implementation with limited opportunity for their input.[12] The learning cycle model is distinct from other cyclical and quality improvement processes and also requires the development of institutional infrastructure to support ongoing and concurrent learning cycles.[5]

Within the learning cycle model, data visualization dashboards can be conceptualized as a means of transforming practice data into shareable knowledge. Dashboards can be especially helpful to monitor changes in clinically important patient behaviors or biomarkers that are difficult to observe, and to communicate these changes with relevant stakeholders. In inpatient rehabilitation settings, progress is generally measured by functional outcomes such as bathing, dressing, standing up from sitting, and walking[13,14]; all of these are easily observable to both healthcare providers and patients. A special case in rehabilitation is patients with disorders of consciousness (DoC) following brain injury. By definition, patients in DoC states may not show consistent, meaningful responses to stimuli or commands and cannot engage in basic self-care and mobility activities that are frequently the target in rehabilitation interventions. Progress is often challenging to identify and reflexive responses can be mistaken for purposeful actions.[15,16] Patients in states of disordered consciousness cannot speak on their own behalf.[17] Their family members or care partners frequently need additional context and information to understand their condition and whether improvement is occurring. Communication between family members and rehabilitation providers is often challenging, as each may interpret observed patient behaviors differently.[15] Third-party payors also may require additional information about a patient's recovery when making decisions about continued payment for rehabilitation services and level of care intensity. Thus, a dashboard showing key metrics to assess level of consciousness and recovery trajectory for patients with DoC can be extremely beneficial for monitoring progress, informing treatment decisions, and enhancing communication.

A team from Spaulding Rehabilitation Hospital in Boston, Massachusetts, developed such a dashboard in 2012 by building a template in Microsoft Excel called the DoC e-Profile to support their specialized DoC rehabilitation program.[18] The DoC program is housed in both an inpatient rehabilitation facility (IRF) and long-term acute care hospital (LTACH), depending on each patient's care needs. Data from a set of standardized assessments were entered into designated sections of the spreadsheet and the template produced corresponding graphs. Although clinicians and researchers at Spaulding reported benefits of using the e-Profile including enhanced communication with patients' families, the process of manually entering assessment data—which duplicated information that was already entered in the EHR system—became untenable after transition to a new EHR system, and its use was discontinued.

1.1 | Research interest

In 2023, Spaulding set out to develop a data visualization tool similar to the DoC e-Profile that would provide the relevant information about individual patients with DoC by using standardized assessment data directly from the EHR and enable aggregation of data across patients. We used an LHS approach for this initiative, gathering input from multiple user groups to ensure their various needs and preferences were identified and addressed with the new design, and using an iterative design approach.[5] The purpose of this study was to

identify, compare, and contrast the informational needs of potential user groups for a visualization tool to show progress for adults with DoC in an inpatient rehabilitation setting. At the start of the project, we confirmed that patient assessment data were consistently captured in discrete fields within the EHR and could be extracted for analysis and visualization (performance to data segment of the continuous learning cycle); therefore, our main focus was on translating this data into knowledge, with the longer-term goal of applying this knowledge to enhance performance.

2 | METHODS

The decision to use an LHS approach for dashboard development required an investment of time and resources, as well as a commitment to utilizing the input gathered in the resulting design.[5,6] Because there was a previous iteration of the dashboard in the form of the e-Profile,[18] this initiative could have been approached as an information technology project, with technicians re-creating the e-Profile on a different technology platform. Instead, Spaulding sought to use an LHS approach to address all user needs, without having a predetermined product. The institutional review board of Spaulding Rehabilitation Hospital determined that this needs assessment did not meet the criteria for human subjects research (IRB ID #671); thus, informed consent was not required.

2.1 | Needs assessment

We conducted a qualitative descriptive study using informational interviews with a convenience sample of key informants including (a) rehabilitation therapists at the health system, (b) family members of patients with DoC, and (c) liaisons for third-party payors.[19–21]

A common set of guiding questions was used for all interviews. Topics included the following: (1) What information does the user need to monitor progress for patients with DoC? (2) What is important for families to understand about changes in patients with DoC? (3) Are there specific assessments that are important to monitor progress? Screen shots of the Excel-based e-Profile tool were shared during each interview to solicit feedback about what features participants liked or disliked, and the ease of understanding findings. Interviews were conducted (AMC) via Zoom and typically lasted approximately 30 min. Data were recorded using contemporaneous written notes. We used contemporaneous notes rather than audio recordings to qualify as a quality improvement project per IRB standards. This approach made it easier to accommodate participation of different team members, particularly through the iterative design stage. No protected health information or personally identifiable information was collected. Interview results were discussed during regular project team meetings that included clinician-researchers at Spaulding.

Data analysis was descriptive. We used matrix analysis to summarize interview data, identify commonalities across user categories, and contrast differing priorities.[22] In the matrix, the interview questions served as column headers. Content from each interview was placed under each relevant question. Once all interviews were entered in the matrix, we identified similarities and differences in responses by user category (rehabilitation therapist, family member, or payor representative). We used a thematic survey approach to organize the findings using the following categories: (a) how the dashboard will be used; (b) content to be displayed; (c) organization and design of content; and (d) technical requirements.[23] Category definition and assignment of content to categories was initially conducted by the first author, then refined after review and input from the second author.

2.2 | Iterative design and feedback process

A synthesis of the information interview findings was shared with interdisciplinary rehabilitation teams at Spaulding during two in-service presentations (one at each location involved in the DoC rehabilitation program). There were opportunities for attendees to ask questions and make suggestions about the initial iteration of the dashboard during this time. We then worked with the informatics team to identify the technology options to support the dashboard and created a prototype version. We selected Tableau because it was already in use within the healthcare system. Tableau can access data directly from the EHR, and the informatics team had the skill set to design a new dashboard.[24] Once the prototype was completed, a second round of meetings was held with the interdisciplinary rehabilitation teams to obtain additional design input. The input received from the rehabilitation team was used to refine the overall layout and was shared with the family members who participated in the information interviews. We continued regular meetings between the research team, informatics personnel, and clinical teams for quality and functionality checks until the clinical team leaders were satisfied with the design.

3 | RESULTS

3.1 | Needs assessment

Seven people participated in informational interviews ($n = 2$ rehabilitation therapists; $n = 2$ family members; $n = 3$ third-party payor representatives). Among the therapists, one was an occupational therapist and one was a physical therapist. Family members were both parents who had been caregivers for their adult children with traumatic DoC. Third-party payor representatives included a lead case manager, a government affairs officer for a healthcare organization, and a chief medical officer at a company that handles workers' compensation claims.

3.1.1 | Dashboard usage

Clinicians identified communication with patients' families/care partners as their primary intended use for the dashboard. They described

how it would be easy to show assessment results in a visual format rather than only giving a verbal explanation. Clinicians indicated a visual depiction of assessment results over time that would be helpful for showing small gains, or highlight that no improvements had occurred. They also suggested that a dashboard would help clinical teams "be on the same page" about whether a patient was getting better, worse, or not changing. This input was incorporated into the initial dashboard design by using line graphs for key assessments on the main screen.

Both family caregiver participants indicated that a visual tool showing their adult child's assessments over time would have been helpful for understanding "where things stand overall." One parent described how "when they tell me the scores, it doesn't mean anything," but she could understand a graph in which she could "see for herself." A dashboard would help them to understand what was working. In response to seeing the example Excel-based e-Profile, one parent offered an interpretation of the graph with an upward trajectory by observing, "What are we doing to make that go up? Let's continue to do that."

Representatives from third-party payors stated two possible uses for a DoC dashboard. First, they liked the idea of being able to house DoC-specific measures in a single place because patients with DoC often do not show progress on most traditional functional measures (e.g., Section GG of the IRF-PAI).[13] The second use was to help determine whether the patient was in the appropriate treatment setting and receiving the appropriate level of care. To meet this need, the initial dashboard design included both LTACH and IRF sites within Spaulding for a single patient, if they had transferred from one to the other.

3.1.2 | Dashboard content

Project team members and rehabilitation therapists identified the assessments that are considered the "core metrics" for the DoC program as priorities for inclusion in the dashboard. These measures include the Coma Recovery Scale-Revised (CRS-R), a standardized measure of neurobehavioral function[25–27]; Limb Movement Protocol which tracks the recovery of functional movement[28]; Disability Rating Scale assessing recovery following coma to return to community[29]; and the Confusion Assessment Protocol tracks changes in seven symptoms associated with post-traumatic confessional state.[30,31] Clinicians emphasized that the language should be simple and interpretable by all users, regardless of whether they have relevant clinical training. Third-party payor representatives did not express any preference for assessments. Family members reported greater interest in seeing the positive, negative, or unchanging direction of the patient's trajectory over time rather than any specific measurement.

As one family member put it, "I need the facts." Family members also stated that it would be helpful to include guidance about the meaning of assessment data. For example, the CRS-R is used to establish a DoC diagnosis (vegetative state [VS], aka unresponsive wakefulness syndrome [UWS], minimally conscious state [MCS], or emerging [eMCS]). Family members indicated that they wanted to see the connection of the assessment score with the DoC diagnosis terms. They stressed the importance of having information that was specific to the patient. As one of the family members stated, "When [my son] was admitted to rehab, they gave me this big binder. I'm sure there was good information in there, but I don't think I ever opened it. I wanted to know about him." We added brief text definitions of the DoC diagnostic states to the dashboard for easy reference against the assessment results.

Rehabilitation stays for patients with DoC are often interrupted by medical complications, many of which require temporary transfers to acute facilities.[32] Therapists want to be able to see an entire episode of care from first admission to final discharge from rehabilitation, across any interruptions or transfers. We addressed this request in the initial dashboard design in how an episode of care was designated. Payor representatives were interested in showing data that would help them better evaluate patients' prognoses for recovery. One specific piece of information requested was the time elapsed between injury and rehabilitation admission, to easily assess the patient's temporal progress.

3.1.3 | Organization and design

Clinicians expressed that they would like to see all of the patient-level core metrics for the DoC program on the same page to support clinical reasoning. Having everything in the same place would make it "easy to understand" what was happening with a patient and how they were changing (or not) over time. They also suggested the use of different colors for each assessment to facilitate identification. In contrast, a family member participant suggested using what she called a "flashcard approach," showing one assessment at a time to make it easier to absorb the information. Family members also suggested a visual they could understand on their own and a simple design with supporting information about how to interpret assessment data. We balanced these needs by putting the key assessments on the main dashboard page, and giving the option to click on a single assessment to review in more detail.

Although payors expressed interest in seeing individual trajectories, they said the information would be more helpful if patient progress were contextualized against normative recovery patterns, if available. Situating individual information against population averages would support their evaluation about whether a person was likely to continue to recover. Family members also stated they would like information about what is "normal" to have more context for what they were observing in the patient as a comparison. Normative data about recovery patterns is not currently available; however, we will consider adding this information in the future.

3.1.4 | Technical requirements

Clinicians requested that using a visualization tool not require additional data input or work by the rehabilitation staff. Because all core

metrics are templated in the EHR system, they requested using the data that are already being inputted as part of the standard of care in the DoC program. At the same time, their previous experience with the report building tools in the EHR was that they were not user-friendly and were "too time consuming." They expressed openness to learning about more advanced or different options within the EHR if they could achieve their desired results. The clinicians also wanted to have a plan for how to request changes or updates to the dashboard after implementation to improve functionality or meet emergent needs. At the institutional level, the goal was to make use of existing technology within or connected to the EHR system rather than purchase or build new software. Figure 1 presents a schematic of how clinician participants described the ideal data flow from clinical encounter through dashboard use. The figure shows the ideal pathway that data collected and recorded during routine care processes would be extracted and then available to use for communication between different interested parties.

Family members, on the other hand, wanted to have their own copy of any assessment data they were shown. "I would want it on paper so I can go back over it" rather than only see something on a screen. Payor representatives often had direct access to EHR systems for viewing patient records, so they expressed a preference for being able to retrieve visual outputs there.

3.2 | Iterative design and feedback process

During the in-service meetings with the interdisciplinary clinical teams, most of the questions and comments were related to the implementation of the dashboard within their existing workflow. For example, attendees asked how they would access the dashboard, whether they would have the capability to capture "snapshots" of the dashboard, and if they could add snapshots to the EHR. We agreed to create a "user guide" document with step-by-step instructions to address these types of questions to support dashboard implementation.

In subsequent meetings with the interdisciplinary rehabilitation team, we shared the prototype dashboard and received feedback about content, layout, and functionality. The original prototype had nine panels on the home screen that showed the four core assessments identified in the needs assessment process (CRS-R, Disability Rating Scale, Limb Movement Protocol, and Confusion Assessment Protocol), plus detail for individual items of each. During these meetings, the home screen was modified to show only four panels with the four core assessments, and detail on individual items was accessed by clicking on the main assessment (Figure 2).

The revised version with rehabilitation team input was shared with family members who participated in the needs assessment in a separate meeting. They favored the visual display being "simple" and that, with line graphs representing the scores for three of the four metrics, "you can see what direction it's going." When asked whether they would still prefer to have a paper version or access a snapshot through a patient portal, they recommended that family members always be given the option of a paper printout, because not everyone will be familiar with how to navigate a patient portal and they "may not want to use their energy to learn now when they are dealing with so many other things."

Our team including researchers, informatics, and clinical team leads continued to hold regular meetings to conduct quality and functionality checks. The draft dashboard was moved into production in Tableau. We created the user guide document to support clinical implementation that included a process for requesting modifications in the future.

4 | DISCUSSION

The purpose of this qualitative descriptive study was to identify the information needs of distinct user groups for a dashboard to monitor recovery for patients with DoC after severe brain injury. Using an LHS approach, we approached the creation of a data visualization dashboard as a process for translating data to knowledge. Rehabilitation

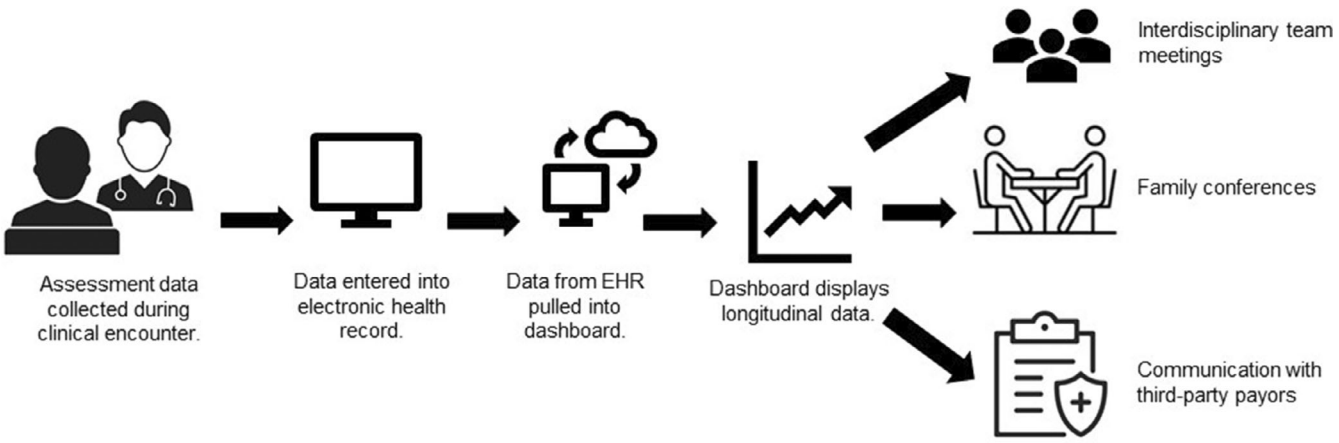

FIGURE 1 Optimal flow of assessment data from clinical encounter to dashboard use.

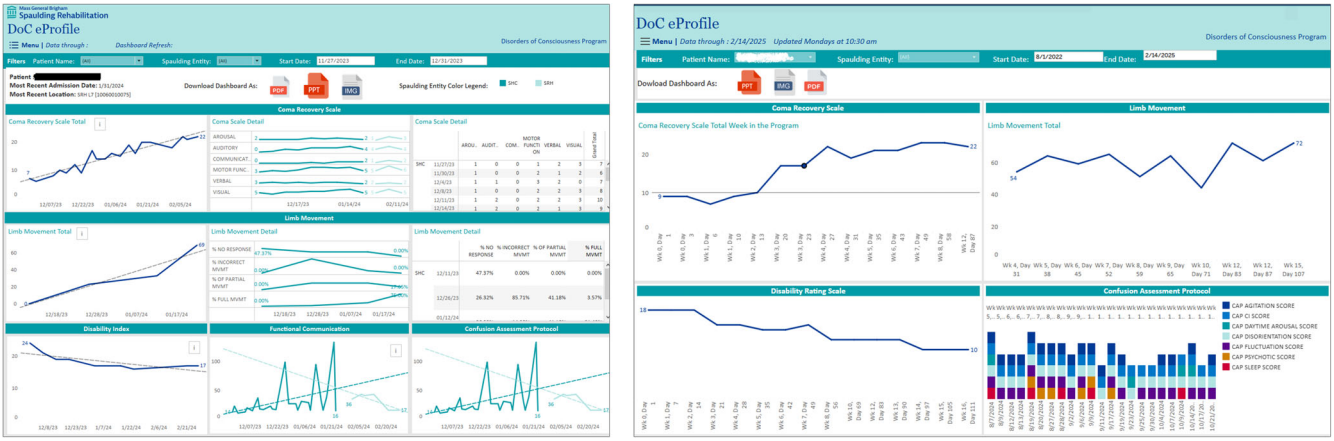

FIGURE 2 Comparison of dashboard design from original draft (left panel) to final version after incorporating user feedback (right panel).

clinicians, family members of patients with DoC, and representatives of third-party payors participated in the study. Although these three groups desired similar content for a dashboard, they each expressed the need for different contextual information with varying design preferences. The iterative design process allowed the primary end users—interdisciplinary rehabilitation teams and family members of patients with DoC—opportunities to directly influence the dashboard content and layout. Our findings highlight the value of applying the continuous learning cycle model to approach a problem without certainty of its solution, engaging a team with multiple user perspectives, and working within existing institutional infrastructure and technology architecture to identify user needs and generate a data visualization dashboard.

A key benefit of this LHS approach was the ability to "embrace uncertainty" about the end product and proactively design the dashboard to account for many of the user needs.[5] This approach increased the likelihood of successful implementation and adoption of the dashboard. Engagement of varied stakeholders uncovered both similarities and differences in needs and preferences of different user groups. Prior to initiating the design and building of the dashboard, there were conversations about whether a single product could simultaneously address the identified needs or if multiple products may be necessary. For example, the contrast between clinicians wanting to see results of multiple assessments simultaneously and family members requesting the one-at-a-time "flashcard approach" might require different end products or dashboard customization options. Team contributions to the iterative design process helped to bridge these different needs; feedback to date suggests the process was successful.

The differences in roles of each user group in supporting the rehabilitation of patients with DoC were apparent in how they preferred information to be presented. For instance, payor representatives were the only users who expressed an interest in seeing individual progress against normative data so that they could make inferences about the probability and magnitude of further recovery. Probability of recovery was the rationale for requesting the context of the injury date, as longer time since injury has been shown in previous studies as a risk for poorer rehabilitation outcomes.[16,33–35] It is conceivable that such information might be used by payors to limit or deny care. Although clinical teams and family members of patients may also find such information beneficial, they prioritized details about the individual patient rather than broader recovery patterns.

There is evidence of third-party payors using patient data to inform decisions about coverage and payment of care[36,37]; however, we are not aware of any research about how payors use individual patient assessment data for decisions about rehabilitation service coverage. This topic is a potential avenue for future research. The standardized functional assessment items mandated by the Centers for Medicare and Medicaid Services for post-acute inpatient rehabilitation are not appropriate for adults with DoC, as they require the ability to follow commands and attempt self-care and mobility tasks.[13] A chronological visual display of DoC core metrics for a specific patient may better inform decisions about continued payment for inpatient rehabilitation or discussions about transfer to a different level of care.

Although our goal was to understand different user needs independent of particular technology tools, technical capabilities and limitations of existing institutional resources are critical to bringing the dashboard to fruition in the future. During the data collection and analysis period, we began engaging with Spaulding's clinical informatics specialist (JEG) to identify the existing data visualization tools available within the healthcare system and how well they would support the user needs and preferences. Early involvement of the informaticist in the process has been critical for selecting the best tool and locating the appropriate data elements within the EHR. It also helped to establish a pathway to utilize existing institutional infrastructure, ongoing maintenance, and improvement, as well as to support future data visualization projects.

4.1 | Limitations

This qualitative descriptive needs assessment used a convenience sample, recruiting from the study team's professional network. It is possible that a broader, more diverse sample would have had different or additional perspectives on each aspect of the data visualization

tool. Applications of an LHS approach for any initiative would typically include patient participants. Patients with DoC, whose care is intended to be supported by our data visualization tool, cannot effectively communicate their wants and needs. Thus, we included input from family members of patients with DoC to represent the patient perspective. There may be other visualization tools that might have produced dashboards with different graphs and display components; however, we chose to limit our options to tools with few implementation barriers, hence the selection of Tableau.

5 | CONCLUSIONS

Data visualization has great potential to support information sharing and communication among different stakeholders about individual patient rehabilitation progress. In the unique case of patients with DoC, who cannot speak for themselves, effective communication among rehabilitation clinicians, family members or care partners, and third-party payors is critical for optimal care. Although these user groups need similar information, how they use and interpret it differs. An LHS approach to identify similar and different perspectives about usage, content, organization, and technical requirements across user groups can inform the design of effective data visualization tools. Future work should include the development of best practices and procedures for user-engaged design of data visualization tools, as well as effective strategies for balancing competing preferences.

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CONFLICT OF INTEREST STATEMENT

Alison Cogan received research funding from the Southern California Clinical and Translational Science Institute (KL2 award) and the US Department of Defense. Shonali Gaudino and James Green are employees of Spaulding Rehabilitation. Lewis Kazis is a consultant to Mediwound. Mary Slavin serves as the Director of the Dissemination Core and Co-Director of Didactic Activities for LeaRRn. Jeffrey Schneider has received research funding from National Institute for Independent Living and Rehabilitation Research (NIDILRR) and the US Department of Defense. Joseph Giacino has received research funding from NIDILRR and occasionally receives honoraria for Coma Recovery Scale-Revised training seminars conducted for academic and healthcare institutions. All authors declare that they have no conflicts of interest related to this work.

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